

Deepak Chetankumar Jain

Data Analyst

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PROFESSIONAL SUMMARY:

- Data Analyst with 4+ years of extensive experience in healthcare and financial sectors, adept at transforming complex datasets into actionable insights to drive strategic outcomes.
- Proficient in developing Python-based automation frameworks and architecting scalable ETL pipelines using PySpark, Pandas and NumPy on Azure and AWS platforms, significantly reducing manual efforts and enhancing data processing efficiency.
- Proven expertise in implementing advanced machine learning models, including Logistic Regression, XGBoost and LightGBM, to improve patient risk stratification and achieve high precision in fraud detection.
- Skilled in optimizing data retrieval with complex T-SQL and PL/SQL queries across Oracle, SQL Server, Snowflake and Teradata and expediting critical reporting through dynamic Tableau and Power BI dashboards.
- Committed to ensuring stringent regulatory compliance (HIPAA, HITECH, RBI, Basel III) and maintaining robust data integrity, consistently delivering improvements in operational efficiency, data-driven decision-making and risk mitigation.

TECHNICAL SKILLS:

Programming Languages:	Python, R, SQL (T-SQL, PL/SQL), VBA
Data Analytics Libraries:	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost, LightGBM, tidyverse, dplyr, ggplot2
Data Visualization:	Tableau, Power BI, Microsoft Excel (PivotTables, Power Query, VBA macros, dynamic charts, slicers)
Data Engineering & ETL:	PySpark, SSIS, Informatica PowerCenter, Azure Data Factory, Mirth Connect, Power Query, Apache Spark
Cloud Platforms:	Microsoft Azure (Azure Databricks, Azure Data Factory), AWS (S3, Redshift)
Databases:	Snowflake, SQL Server, Oracle, Teradata, PostgreSQL
Healthcare Data Standards:	EDI 837/835, HL7 v2.x, FHIR, ICD-10, CPT, HCPCS, HEDIS
BI & Reporting Tools:	Tableau, Power BI, Microsoft Excel, SQL Reporting Services (SSRS)
Statistical Analysis:	A/B Testing, Time-Series Analysis, Cohort Analysis, Root Cause Analysis (RCA), Feature Engineering, Regression Analysis
Data Governance & Security:	HIPAA, HITECH, KYC, AML, Basel III, Data Encryption, De-identification, Audit Trails
Version Control & DevOps:	Git, GitHub
EHR & EMR Systems:	Epic Systems, Allscripts
Workflow Automation:	Python Automation Scripts, VBA Macros, Mirth Connect Channels
Methodologies:	Agile, Scrum, Waterfall
Other Skills:	Ggplot, Jupyter Notebook, decision tree, linear regression, Data Storytelling, Data Wrangling, Regression analysis, A/B testing, Data Modeling, Data Cleaning, Data Transformation, Data Visualization, Data warehousing
Operating Systems:	Windows, Linux, Mac iOS

PROFESSIONAL EXPERIENCE:

Cardinal Health – NJ

September 2023 – Present

Data Analyst

- Reduced healthcare claim rejection rate by 31% by developing a Python-based automation framework to validate EDI 837/835 transactions against payer-specific rules and CMS standards.
- Enhanced patient risk stratification accuracy by 22% through implementing logistic regression and XGBoost models, leading to better care coordination for high-risk patients under value-based care programs.
- Streamlined data preparation processes by 45% by defining scalable data transformation and validation logic for claims and EHR data in PySpark.

- Optimized reporting efficiency by 60% through pre-aggregating data in Snowflake and configuring Tableau incremental refresh and materialized views.
- Enabled real-time access to clinical data by integrating Epic Systems data via Azure Data Factory pipelines and FHIR APIs for timely analysis.
- Streamlined HL7 lab message ingestion workflow by 40% through Mirth Connect channel configurations that parsed, cleansed and mapped HL7 v2.x messages into normalized relational structures.
- Identified chronic disease utilization patterns by performing time-series and cohort-based data mining in R using tidyverse and dplyr packages, informing preventive care strategies.
- Wrote complex T-SQL and PL/SQL queries to join and normalize high-volume datasets across Oracle and SQL Server from EHR (Epic/Allscripts), claims, pharmacy and EDI repositories.
- Designed Excel-based KPI dashboards using VBA macros, Power Query, dynamic charts and slicers to track provider efficiency, patient throughput and HEDIS quality measure compliance.
- Performed root cause analysis (RCA) on data integrity issues between EHR and payer systems, reconciling encounter records and resolving mismatches within SLA timelines (<72 hours).
- Ensured compliance with HIPAA and HITECH requirements by applying data encryption, de-identification, and audit controls to reporting workflows.
- Collaborated with clinical SMEs, coders and quality analysts to accurately map ICD-10, CPT and HCPCS codes to clinical episodes, ensuring appropriate risk adjustment and claims adjudication.

Capegemini – India
Data Analyst

April 2020 – August 2022

- Optimized Python-driven ETL pipelines utilizing Pandas, NumPy and SQL to extract, transform and load complex loan portfolio datasets from Oracle and Teradata databases, improving data processing efficiency by 25%.
- Deployed dynamic Power BI dashboards with drill-down capabilities to monitor critical risk metrics, enabling real-time tracking of Non-Performing Assets (NPAs) and supporting strategic decision-making for senior management.
- Conducted rigorous A/B testing and statistical analysis to assess the impact of revised credit policies on loan approval rates and default probabilities, reducing risk exposure by 10% through data-driven insights.
- Ensured strict adherence to RBI regulatory frameworks and Basel III norms by embedding automated KYC and AML compliance checks into data workflows, achieving 100% compliance during audits.
- Engineered optimized SQL queries and PL/SQL stored procedures to streamline data retrieval processes for credit risk modeling, reducing query execution time by 30%.
- Leveraged SSIS and Informatica PowerCenter to orchestrate seamless ETL workflows, integrating heterogeneous data sources into a unified data mart for accurate and timely financial reporting.
- Designed and implemented advanced fraud detection models using LightGBM and feature engineering, achieving a precision rate of 92% in identifying anomalous transactions across high-volume datasets.
- Created sophisticated Excel-based reports using PivotTables, Power Query and VBA macros to facilitate ad-hoc loan portfolio analysis, enabling rapid response to stakeholder queries.
- Deployed scalable data pipelines on AWS S3 and Redshift, incorporating encryption and access controls to ensure secure storage and compliance with data governance standards.
- Improved fraud detection accuracy by 18% by fine-tuning LightGBM models with hyperparameter optimization, significantly mitigating financial losses from fraudulent activities.

EDUCATION:

Master of Computer Science - Stevens institute of technology, Hoboken, New Jersey, USA

Bachelor of Computer Engineering - Terna engineering college, Navi mumbai, Maharashtra, India